19. Define moment generating function.

20. State the Characteristic function for a continuous random variable.

21. If a random variable has the following probability distribution

| $X$ | o | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X=x)$ | $3a$ | $4a$ | $6a$ | $7a$ | $8a$ |

find the value of '$a$'.

22. Verify whether the following is a p.d.f. $f(x) = 5x^4$, $0<x<1$?

23. A random variable $X$ has $E(X)=\frac{1}{2}$, $E(X^2)=\frac{1}{2}$ find its standard deviation

## IV. Answer in brief:

24. Write down the properties of distribution function.

25. Find the probability distribution of 6's in throwing 3 dice once.

26. A box contains 6 red and 4 white balls. Three balls are drown at random obtain the probability distribution of the number of white balls drawn.

27. A random variable $X$ has the density function ) $f(x) = \begin{cases} \frac{1}{4} & -2 < x < 2 \\ 0 & elsewhere \end{cases}$
Then find (i) $P(-1 < X < 2)$ and (2) $P(X>1)$.

28. A game is played with single fair die. A player wins ₹ 20 if a 2 turns up ₹ 40 if a 4 turns up. Looses ₹ 30 if a 6 turns up. While he neither win nor looses If any other face turns up. Find his expectation of gain.

## IV. Calculate the following:

29. In a continuous distribution the p.d.f. of $X$ is
$f(x) = \begin{cases} \frac{3x(2-x)}{4} & 0 < x < 2 \\ 0 & elsewhere \end{cases}$ Find the mean and variance

30. If the *pdf* of a continuous random variable $x$ is given by
$f(x) = \begin{cases} \frac{k(1-x^2)}{4} & 0 < x < 1 \\ 0 & elsewhere \end{cases}$
Find $K$ and $E(X)$ :

31. Two unbiased dice are thrown together at random. Find the expected value of a total number of points shown up.